

DATA QUALITY REPORT

StreamFlix Content Analytics

Purpose: Assess the quality of the StreamFlix analytical tables before downstream SQL, EDA and Power BI analysis. The report follows the task-oriented structure of the referenced Data Quality Analysis example, while using the actual checks and results recorded in the supplied Jupyter notebook.

Scope: The notebook inspected six relational tables — Subscribers, Titles, Ratings, Reviews, Watch History and Watchlist — using completeness, uniqueness, validity, consistency and relationship checks.

1. Executive Summary

Overall, the inspected data is structurally usable for analysis, with no duplicate primary-key records detected in the tested tables and no nulls in Ratings, Reviews or Watchlist. The main completeness issues are conditional rather than universally erroneous: churn_date is null for active subscribers, while license_expiry is null for a subset of titles. Several business-rule checks also passed, including subscriber signup/churn date ordering, watch-duration limits and Watchlist-to-Subscriber referential validation.

The most important reporting consideration is semantic interpretation of nulls. A null churn_date for an active subscriber is expected and should not automatically be treated as a missing-data defect. Similarly, missing license_expiry should be investigated according to the title's licensing arrangement before imputation or deletion.

2. Data Inventory & Quality Summary

Table	Volume	Duplicate PKs	Null findings	Assessment
Subscribers	15,000 × 14	0	11,199 churn_date nulls	Conditional nulls; key uniqueness passed
Titles	9,000 × 21	0	1,966 license_expiry nulls	Conditional completeness issue
Ratings	130,000 × 5	0	0	Clean on tested checks
Reviews	110,000 × 7	0	0	Clean on tested checks
Watch History	650,000 × 11	0	0	Business-rule checks passed
Watchlist	65,000 × 5	0	0	Subscriber relationship check passed

3. Data Quality Dimensions Assessed

Dimension	Assessment approach
Completeness	Presence of required values and interpretation of legitimate nulls.
Uniqueness	Duplicate detection on tested primary-key fields.
Validity	Values conform to expected business rules and categories.
Consistency	Related fields agree logically, such as completion percentage and dates.
Referential integrity	Child records point to valid parent identifiers where tested.

4. Key Findings

- **Subscribers:** 15,000 records were inspected. subscriber_id had no duplicates. Of the 15,000 subscribers, 11,199 are active and have null churn_date; 3,801 have a churn date. The notebook found 3,801 valid signup_date < churn_date records and zero invalid date-order records.
- **Titles:** 9,000 records were inspected with no duplicate title_id values. license_expiry contains 1,966 nulls (21.84% of title records), making this the principal title-level completeness issue.
- **Ratings and Reviews:** No duplicate tested IDs and no null values were found. Ratings use rating_id as the tested key; Reviews use review_id. Review sentiment values were restricted to Positive, Negative and Neutral in the notebook.

DATA QUALITY REPORT — DETAILED ASSESSMENT

5. Detailed Validation Results

Validation	Records	Business rule	Result
Subscriber date logic	3,801 churned records	signup_date < churn_date	Passed; 0 invalid records
Active subscriber status	11,199 active records	churn_date should be null	Passed; 11,199/11,199 satisfy rule
Watch duration	650,000 watch records	watch_duration_min ≤ content_duration_min	Passed; 0 violations
Completion %	650,000 watch records	watch_duration/content_duration × 100	Values align after rounding
Watchlist subscriber link	65,000 records	subscriber_id exists in Subscribers	Passed; 65,000 valid, 0 invalid
Review sentiment	110,000 reviews	Expected category values	Positive / Negative / Neutral observed

6. Issues Requiring Attention

Date Data Type Validation

During the data quality assessment, several date-related columns were initially identified with an object data type instead of a proper date/datetime data type. This could affect date-based sorting, filtering, duration calculations, time-series analysis, and reporting.

Action Taken: The affected date columns were converted from object to datetime using pandas `pd.to_datetime()`. After conversion, the columns could be reliably used for date comparisons, chronological analysis, time-based calculations, and Power BI date-based reporting.

Status: Resolved after data type conversion.

- 1) Conditional nulls in Subscribers — churn_date. 11,199 records are null. This is consistent with the active-status rule observed in the notebook, so these values should be retained as null rather than imputed. The report should explicitly document this as an expected business condition.
- 2) Missing license expiry dates — Titles. 1,966 of 9,000 titles have no license_expiry (21.84%). The business meaning should be confirmed: the value may be unavailable, not applicable, perpetual, or missing from the source. The field should not be filled with an arbitrary date.
- 3) Date-type standardisation. The notebook identifies date fields such as rating_date and added_date that were initially represented as object/string values and converts some of them to datetime. A standard date type should be enforced consistently before time-based analysis.
- 4) Validation coverage. The notebook performs strong structural checks, but additional domain checks should be added before production reporting — for example, rating range validation, non-negative financial measures, valid age/household ranges, valid title references in Ratings/Reviews/Watch History, and uniqueness of relevant business keys.